

Heavy Hitters

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CS 663

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Motivation

Stream of items (IP addresses, search queries, shopping items, etc.)

Example: Twitter stream of hashtags

#aot, #covid, #SP500, #bitcoin, #dogecoin, #aot, #covid,
#covid, #aot, #onepiece, #covid, #aot, . . .

Goal: Find frequent items

Motivation

Storing the entire stream in RAM for counting is expensive.

Keeping a counter for each item encountered is expensive. The number of items gets larger and larger over time.

Misra-Gries sketch

Suppose we have a stream of elements

$$x_1, x_2, \dots, x_m \in \{1, 2, \dots, n\} = [n].$$

Find all elements that appear more than m/k times.

Misra-Gries sketch

a, a, b, c, a, a, a, d, a, a ($m = 10$)

For $k = 3$, return a since $\text{count}(a) > 10/3$.

Misra-Gries sketch

Algorithm 1: Misra-Gries

Initialize an empty hash map H (i.e., dictionary)

for $i = 1$ to m do

 if $x_i \in H$ then

 // If x_i is in one of the counters, increment the counter.

$H[x_i] \leftarrow H[x_i] + 1$

 else if $|H| < k - 1$ then

 // If not, and # counters $< k - 1$, open a new counter for x .

$H[x_i] \leftarrow 1$

 else

 // If there are $k - 1$ counters, decrement every counter by 1. Delete those whose count reaches 0.

 for each $y \in H$ do

$H[y] \leftarrow H[y] - 1$

 if $H[y] = 0$ then

 Remove y from H

return H

Misra-Gries sketch – Example

Detect all elements that appear more than $m/4$ times ($k = 4$, so $k - 1 = 3$ counters).

Stream: a, a, b, c, c, a, a, c, d, e, e, c ($m = 12$).

Step	Item	Action	Counter State
1	a	New counter for a	$H[a]=1$
2	a	Increment a	$H[a]=2$
3	b	New counter for b	$H[a]=2, H[b]=1$
4	c	New counter for c	$H[a]=2, H[b]=1, H[c]=1$
5	c	Increment c	$H[a]=2, H[b]=1, H[c]=2$
6	a	Increment a	$H[a]=3, H[b]=1, H[c]=2$

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2	a	Increment <i>a</i>	$H[a]=2$
3	b	New counter for <i>b</i>	$H[a]=2, H[b]=1$
4	c	New counter for <i>c</i>	$H[a]=2, H[b]=1, H[c]=1$
5	c	Increment <i>c</i>	$H[a]=2, H[b]=1, H[c]=2$
6	a	Increment <i>a</i>	$H[a]=3, H[b]=1, H[c]=2$

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2	a	Increment a	$H[a]=2$
3	b	New counter for b	$H[a]=2, H[b]=1$
4	c	New counter for c	$H[a]=2, H[b]=1, H[c]=1$
5	c	Increment c	$H[a]=2, H[b]=1, H[c]=2$
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Misra-Gries sketch – Example (cont.)

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Step	Item	Action	Counter State
7	a	Increment a	$H[a]=4, H[b]=1, H[c]=2$
8	c	Increment c	$H[a]=4, H[b]=1, H[c]=3$
9	d	3 counters full, decrement all	$H[a]=3, H[c]=2$ (b removed)
10	e	New counter for e	$H[a]=3, H[c]=2, H[e]=1$
11	e	Increment e	$H[a]=3, H[c]=2, H[e]=2$
12	c	Increment c	$H[a]=3, H[c]=3, H[e]=2$

$$H[a] = 3$$

$$H[c] = 3$$

$$H[e] = 2$$

All heavy-hitters (a and c , each appearing $> m/4 = 3$ times) remain.
Some non-heavy hitters (e) might remain.

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Misra-Gries sketch

Theorem: $\text{count}(x) > m/k$, then $H[x]$ exists at the end.

Misra-Gries sketch

Think of each element as a fruit being stored in a basket.

apple, apple, banana, citrus, citrus, apple, citrus, apple, coconut, apple, coconut, kiwi.

For $k = 4$, after the first 8 elements: we have 3 baskets:

Basket 1: *apple, apple, apple, apple*

Basket 2: *banana*

Basket 3: *citrus, citrus, citrus*

coconut arrives

Remove one element from each basket:

Basket 1: *apple, apple, apple, ~~apple~~*

Basket 2: *~~banana~~*

Basket 3: *citrus, citrus, ~~citrus~~*

4 pieces (including the newly arrived coconut) got deleted.

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Misra-Gries sketch

When we see X , we put X in a basket if
there is an empty basket, or
some current basket contains X

We throw X away if all baskets are full and none contains X .

Every time we throw away a copy of X , we throw away $k - 1$ copies of other items in the current counters. Hence, we lost at most m/k counts of X .

$$H[x] \geq \text{count}(x) - m/k$$

Thus, if $\text{count}(x) > m/k$, $H[x]$ must exist at the end.

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Count-Min sketch

What happens if we allow deletions?

Example: Amazon stream of shopping items. Sometimes an item is returned or removed from the cart.

Misra-Gries does not support deletions. However, we can use Count-Min sketch (see textbook).